Project Plan

Academic Skills Practice App

Document Information

Item	Details
Project Name	Academic Skills Practice App
Version	1.0
Date	2026/3/6
Prepared By	BOT_0
Project Duration	6 Weeks (4 Sprints)

1. Executive Summary

1.1 Project Overview

This project aims to develop a mobile application that helps incoming students practice academic English skills before starting their degree. The app will provide comprehensive coverage of all four language skills - **Listening, Speaking, Reading, and Writing** - plus academic vocabulary, through interactive, bite-sized practice to build confidence and support self-paced learning.

1.2 Project Objectives

1. Build a functional mobile application covering all academic English skills.
2. Provide practice modules for Listening, Speaking (Presentation), Reading, and Writing.
3. Enable users to track their learning progress across all skill areas.
4. Deliver working software through iterative Agile development with rotating leadership.
5. Complete the project within 6 weeks following Scrum methodology.

1.3 Success Criteria

Criteria	Measurement
Skill Coverage	All 4 skills (Listening, Speaking, Reading, Writing) + Vocabulary

Criteria	Measurement
Working Software	All MVP features functional and tested
User Experience	Clean, intuitive interface
Process Adherence	Agile practices followed; leadership roles rotated
On-time Delivery	All milestones met within 6-week schedule

2. Team Organisation

2.1 Team Members

Name	Technical Expertise
Yanbin Xu	Backend Development ()
Yanshuo Liu	Frontend Development ()
Junfan Zhou	Frontend Development ()
Yupei Yang	Backend Development ()
Ruoqi Hu	Full-stack Development ()
Weike Jin	Frontend Development ()
Chenxi Huang	Backend Development ()

2.2 Role Rotation Schedule

Each Sprint will have different Product Owner and Scrum Master to ensure all team members gain leadership experience.

Sprint	Week	Product Owner	Scrum Master	Development Team
Sprint 1	Week 3	**Yanbin Xu**	**Yanshuo Liu**	Junfan Zhou, Yupei Yang, Ruoqi Hu, Weike Jin, Chenxi Huang
Sprint 2	Week 4	**Junfan Zhou**	**Yupei Yang**	Yanbin Xu, Yanshuo Liu, Ruoqi Hu, Weike Jin, Chenxi Huang
Sprint 3	Week 5	**Ruoqi Hu**	**Weike Jin**	Yanbin Xu, Yanshuo Liu, Junfan Zhou, Yupei Yang, Chenxi Huang

Sprint	Week	Product Owner	Scrum Master	Development Team
Sprint 4	Week 6	**Chenxi Huang**	**Yanbin Xu**	Yanshuo Liu, Junfan Zhou, Yupei Yang, Ruoqi Hu, Weike Jin

2.3 Role Responsibilities

Product Owner (Rotating)

- Define and prioritise the sprint backlog
- Clarify requirements and acceptance criteria
- Communicate with the client
- Accept or reject completed work at the Sprint Review
- Ensure the team builds valuable features

Scrum Master (Rotating)

- Facilitate daily standups, sprint planning, review, and retrospective
- Remove obstacles that block the team
- Ensure Scrum practices are followed
- Coach the team and protect them from distractions
- Track sprint progress

Development Team (All Members)

- Self-organise to complete sprint goals
- Collaborate to deliver working software
- Participate in all Scrum ceremonies
- Support rotating PO and SM
- Continuously improve processes

3. Product Scope

3.1 Academic Skills Coverage

The app will provide comprehensive practice for all academic English skills:

Skill Area	Module	Description
Vocabulary	Academic Vocabulary	Learn and practice academic word

Skill Area	Module	Description
	Practice	lists through quizzes
Listening	Lecture Comprehension	Listen to academic lectures and answer comprehension questions
Speaking	Academic Presentation	Practice presentation skills with prompts and self-recording
Reading	Academic Reading	Read academic texts and answer comprehension questions
Writing	Academic Writing	Practice academic writing with structured prompts and examples

3.2 Feature List by Priority

P0 - Must Have (MVP)

ID	Feature	Skill Area	Description
F001	User Registration/Login	Core	Email-based authentication system
F002	Academic Vocabulary Practice	Vocabulary	Multiple-choice vocabulary exercises
F003	Listening Comprehension	Listening	Audio playback with comprehension questions
F004	Academic Reading Practice	Reading	Reading passages with comprehension questions
F005	Academic Writing Practice	Writing	Writing prompts with example responses
F006	Presentation Practice	Speaking	Presentation prompts with tips and self-recording
F007	Progress Tracking	Core	Visual display of progress across all skill areas

P1 - Should Have

ID	Feature	Description
F008	Daily Practice Reminders	Push notifications for practice
F009	Wrong Answer Review	Review incorrect answers for each skill
F010	Detailed Statistics	Performance analytics by skill area

P2 - Nice to Have

ID	Feature	Description
F011	Difficulty Levels	Adaptive difficulty based on performance
F012	Achievement Badges	Gamification elements
F013	Offline Mode	Download content for offline use

3.3 Out of Scope

- Real-time speech recognition and scoring
- AI-powered writing feedback
- Live tutoring or video conferencing
- Integration with university LMS systems
- Peer-to-peer interaction features

4. Project Schedule

4.1 Overall Timeline (6 Weeks)

Week	Phase	Focus	Key Deliverables	Responsible PO
Week 1	Preparation	Team Formation & Planning	Team agreement, product backlog, role assignments	
Week 2	Preparation	Setup & Design	Dev environment, UI wireframes, architecture, content	
Week 3	Sprint 1	Auth + Vocabulary + Listening	User auth, vocabulary module, listening module	Yanbin Xu
Week 4	Sprint 2	Reading + Writing	Reading module, writing module	Junfan Zhou
Week 5	Sprint 3	Speaking + Progress	Presentation module, comprehensive progress tracking	Ruoqi Hu
Week 6	Sprint 4	Integration & Delivery	Polish, testing, documentation, final demo	Chenxi Huang

5. Preparation Phase (Weeks 1-2)

5.1 Week 1: Team Formation & Planning

Goal: Establish team foundation, assign roles, and create product backlog

Task	Assignee
Form team and confirm members	All
Discuss and assign initial roles	All
Meet with instructor to clarify requirements	Yanbin Xu
Set up GitHub repository with branch strategy	Yupei Yang
Create product backlog with all features	Yanbin Xu
Prioritise user stories for all 5 skill areas	All
Research and select technology stack	Yupei Yang, Junfan Zhou
Draft initial project plan	Yanshuo Liu

5.2 Week 2: Setup & Design

Goal: Complete all preparation including content for all skill modules

Task	Assignee	Estimate
Set up project management tool	Yanshuo Liu	2h
Define coding standards document	Junfan Zhou	2h
Set up development environments (all members)	All	6h
Design system architecture	Yupei Yang	4h
Design database schema (all modules)	Chenxi Huang	4h
Create UI wireframes for all screens	Junfan Zhou	8h
Prepare vocabulary content (100+ words)	Chenxi Huang	4h

Task	Assignee	Estimate
Prepare listening content (5+ audio clips + questions)	Ruoqi Hu	5h
Prepare reading content (5+ passages + questions)	Yanbin Xu	5h
Prepare writing prompts (10+ prompts + examples)	Yanshuo Liu	4h
Prepare speaking/presentation prompts (10+ topics)	Weike Jin	4h
Create API contract documentation	Yupei Yang	3h
Set up CI/CD pipeline	Yupei Yang	3h

Week 2 Total: 56 hours

6. Sprint Plans

6.1 Sprint 1: Authentication + Vocabulary + Listening (Week 3)

Product Owner: Yanbin Xu
Scrum Master: Yanshuo Liu

Sprint Goal: Deliver user authentication, vocabulary practice, and listening comprehension modules

User Stories

US-001: User Registration

```
As a new user
I want to register an account using my email
So that I can save my learning progress

Acceptance Criteria:
- User can enter email and password
- Password requires minimum 8 characters
- Confirmation message upon successful registration
- Error messages for invalid inputs
```

US-002: User Login

```
As a registered user
```

I want to log in to my account
So that I can access my saved progress

Acceptance Criteria:
- User can log in with email and password
- "Remember me" option available
- Error messages for incorrect credentials

US-003: Vocabulary Practice

As a learner
I want to practice academic vocabulary
So that I can expand my academic word knowledge

Acceptance Criteria:
- Each session has 10 multiple-choice questions
- Immediate feedback on each answer
- Session score displayed at the end
- Progress saved to user profile

US-004: Listening Comprehension

As a learner
I want to practice listening comprehension
So that I can improve my lecture understanding

Acceptance Criteria:
- Audio clips play with controls (play/pause/replay)
- Comprehension questions appear after listening
- Feedback provided for answers
- Progress saved to user profile

Task Breakdown

Task ID	Task	Assignee	Estimate
T1-01	Set up database and user schema	Yupei Yang	3h
T1-02	Implement registration API	Yupei Yang	3h
T1-03	Implement login API	Chenxi Huang	3h
T1-04	Design and implement auth screens	Junfan Zhou	5h

Task ID	Task	Assignee	Estimate
T1-05	Implement navigation framework	Weike Jin	3h
T1-06	Create vocabulary database tables	Chenxi Huang	2h
T1-07	Implement vocabulary quiz API	Yupei Yang	4h
T1-08	Implement vocabulary practice screen	Junfan Zhou	5h
T1-09	Implement answer feedback component	Weike Jin	3h
T1-10	Set up audio file storage	Yupei Yang	2h
T1-11	Implement listening API	Chenxi Huang	3h
T1-12	Implement audio player component	Weike Jin	4h
T1-13	Implement listening practice screen	Ruoqi Hu	5h
T1-14	Write unit tests	Ruoqi Hu	3h
T1-15	Integration testing	Weike Jin	2h
T1-16	Sprint Review preparation	Yanbin Xu (PO)	2h

Sprint 1 Total: 52 hours

6.2 Sprint 2: Reading + Writing Modules (Week 4)

Product Owner: Junfan Zhou
Scrum Master: Yupei Yang

Sprint Goal: Deliver reading comprehension and academic writing practice modules

User Stories

US-005: Reading Comprehension

```
As a learner
I want to practice academic reading
So that I can improve my reading comprehension skills

Acceptance Criteria:
- Academic passages displayed clearly
```

- Comprehension questions after reading
- Immediate feedback on answers
- Progress tracked and saved

US-006: Academic Writing Practice

As a learner
I want to practice academic writing
So that I can improve my writing skills

Acceptance Criteria:
- Writing prompts clearly displayed
- Example responses available for reference
- Text input area for user writing
- Tips for academic writing provided
- User can save/submit their response

Task Breakdown

Task ID	Task	Assignee	Estimate
T2-01	Create reading content database	Chenxi Huang	2h
T2-02	Implement reading passages API	Yupei Yang	3h
T2-03	Design reading module UI	Junfan Zhou	3h
T2-04	Implement reading passage display	Yanshuo Liu	4h
T2-05	Implement reading questions component	Weike Jin	4h
T2-06	Implement reading feedback and scoring	Ruoqi Hu	3h
T2-07	Create writing prompts database	Chenxi Huang	2h
T2-08	Implement writing module API	Yanbin Xu	3h
T2-09	Design writing module UI	Junfan Zhou	3h
T2-10	Implement writing prompt display	Yanshuo Liu	3h
T2-11	Implement text input with	Weike Jin	4h

Task ID	Task	Assignee	Estimate
	formatting		
T2-12	Implement example response viewer	Ruoqi Hu	3h
T2-13	Implement save/submit functionality	Yanbin Xu	3h
T2-14	Write unit tests	Ruoqi Hu	3h
T2-15	Integration testing	Weike Jin	2h
T2-16	Sprint Review preparation	Junfan Zhou (PO)	2h

Sprint 2 Total: 47 hours

6.3 Sprint 3: Speaking/Presentation + Progress (Week 5)

Product Owner: Ruoqi Hu
Scrum Master: Weike Jin

Sprint Goal: Deliver presentation practice module and comprehensive progress tracking

User Stories

US-007: Academic Presentation Practice

```
As a learner
I want to practice academic presentations
So that I can build confidence in seminar participation

Acceptance Criteria:
- Presentation topics/prompts displayed
- Preparation tips and structure guidance provided
- Optional self-recording feature
- Key phrases and expressions provided
- Progress tracked
```

US-008: Comprehensive Progress Tracking

```
As a learner
I want to see my progress across all skill areas
So that I can understand my overall improvement

Acceptance Criteria:
```

- Dashboard showing all 5 skill areas
- Progress percentage for each skill
- Total practice time displayed
- Streak (consecutive days) shown
- Visual charts/graphs for progress

Task Breakdown

Task ID	Task	Assignee	Estimate
T3-01	Create presentation prompts database	Chenxi Huang	2h
T3-02	Implement presentation API	Yupei Yang	3h
T3-03	Design presentation module UI	Junfan Zhou	3h
T3-04	Implement prompt and tips display	Yanshuo Liu	4h
T3-05	Implement key phrases component	Yanbin Xu	3h
T3-06	Implement audio recording feature	Yupei Yang	5h
T3-07	Implement recording playback	Weike Jin	3h
T3-08	Design progress dashboard UI	Junfan Zhou	4h
T3-09	Implement progress statistics API	Chenxi Huang	4h
T3-10	Implement skill progress cards	Yanshuo Liu	3h
T3-11	Implement progress charts	Yanbin Xu	4h
T3-12	Implement streak tracking	Ruoqi Hu	3h
T3-13	Write unit tests	Ruoqi Hu	3h
T3-14	Integration testing	Yanshuo Liu	2h
T3-15	Sprint Review preparation	Ruoqi Hu (PO)	2h

Sprint 3 Total: 48 hours

6.4 Sprint 4: Integration & Final Delivery (Week 6)

Product Owner: Chenxi Huang

Scrum Master: Yanbin Xu

Sprint Goal: Complete integration, testing, documentation, and final delivery

Tasks

Task ID	Task	Assignee	Estimate
T4-01	Full system integration testing	Weike Jin	4h
T4-02	Cross-module navigation testing	Ruoqi Hu	3h
T4-03	Bug fixing - Critical issues	All	10h
T4-04	UI polish and consistency	Junfan Zhou	4h
T4-05	Performance optimisation	Yupei Yang	3h
T4-06	Implement daily reminders (P1)	Yanshuo Liu	4h
T4-07	User acceptance testing	Chenxi Huang (PO)	3h
T4-08	Complete user documentation	Chenxi Huang	3h
T4-09	Complete technical documentation	Yupei Yang	3h
T4-10	Final deployment	Yupei Yang	2h
T4-11	Prepare presentation slides	Chenxi Huang (PO)	4h
T4-12	Prepare demo video	Yanshuo Liu	3h
T4-13	Rehearse final presentation	All	2h
T4-14	Package all deliverables	Ruoqi Hu	2h
T4-15	Final Sprint Review	All	2h

Sprint 4 Total: 52 hours

7. Effort Summary

7.1 Role Rotation Summary

Team Member	Sprint 1	Sprint 2	Sprint 3	Sprint 4
Yanbin Xu	**PO**	Developer	Developer	**SM**

Team Member	Sprint 1	Sprint 2	Sprint 3	Sprint 4
Yanshuo Liu	**SM**	Developer	Developer	Developer
Junfan Zhou	Developer	**PO**	Developer	Developer
Yupei Yang	Developer	**SM**	Developer	Developer
Ruoqi Hu	Developer	Developer	**PO**	Developer
Weike Jin	Developer	Developer	**SM**	Developer
Chenxi Huang	Developer	Developer	Developer	**PO**

8. Content Requirements

8.1 Content Preparation Checklist

Skill Area	Content Type	Quantity	Owner	Due
Vocabulary	Academic word list with definitions	100+ words	Chenxi Huang	Week 2
Vocabulary	Multiple choice questions	50+ questions	Chenxi Huang	Week 2
Listening	Audio clips (1-3 min each)	5+ clips	Ruoqi Hu	Week 2
Listening	Comprehension questions	25+ questions	Ruoqi Hu	Week 2
Reading	Academic passages (300-500 words)	5+ passages	Yanbin Xu	Week 2
Reading	Comprehension questions	25+ questions	Yanbin Xu	Week 2
Writing	Writing prompts	10+ prompts	Yanshuo Liu	Week 2
Writing	Example responses	10+ examples	Yanshuo Liu	Week 2
Speaking	Presentation topics	10+ topics	Weike Jin	Week 2
Speaking	Presentation tips & key phrases	20+ phrases	Weike Jin	Week 2

8.2 Content Sources (Suggestions)

- Academic Word List (AWL) by Averil Coxhead
- IELTS/TOEFL practice materials (for format reference)
- University lecture recordings (with permission)

- Open educational resources (OER)
- Team-created original content

9. Risk Management

9.1 Risk Register

ID	Risk	Probability	Impact	Mitigation	Owner
R1	Role rotation causes confusion	Medium	Medium	Clear handover at sprint boundaries; documentation	Current SM
R2	Content not ready in time	Medium	High	Start content prep in Week 1; assign clear owners	Rotating PO
R3	Audio recording feature complex	Medium	Medium	Make it optional; simplify to basic recording	Yupei Yang
R4	Tight timeline (1-week sprints)	High	Medium	Focus strictly on MVP; daily standups	Current SM
R5	Team member unavailable	Medium	High	Cross-training; pair programming; documentation	Current SM
R6	Integration issues	Medium	Medium	Define APIs early; continuous testing	Yupei Yang
R7	Scope restriction	Medium	High	Strict MVP focus; PO controls scope	Rotating PO

9.2 Contingency Plans

If content is not ready:

1. Use placeholder content for development
2. Reduce quantity, maintain quality
3. Prioritise vocabulary and listening (Sprint 1)

If a module is behind schedule:

1. Simplify the module's features
2. Move non-essential features to "nice-to-have"
3. Get help from team members ahead of schedule

If role rotation causes issues:

1. Previous PO/SM provides 30-min handover
2. Maintain a shared "decisions log" document
3. Keep communication channels open

10. Communication Plan

10.1 Sprint Ceremonies (Rotating Facilitators)

Ceremony	Frequency	Duration	Facilitator
Sprint Planning	Start of each sprint	1.5h	Scrum Master (rotating)
Daily Standup	Daily	15 min	Scrum Master (rotating)
Sprint Review	End of each sprint	1h	Product Owner (rotating)
Sprint Retrospective	End of each sprint	30 min	Scrum Master (rotating)

10.2 Handover Protocol (For Role Rotation)

At the end of each sprint, outgoing PO and SM must:

1. **Document all decisions** made during the sprint
2. **Update the product backlog** with current priorities
3. **Brief incoming PO/SM** (30-minute handover meeting)
4. **Transfer any pending issues** or blockers
5. **Share lessons learned** from the sprint

11. Quality Assurance

11.1 Testing Strategy by Module

Module	Test Focus	Responsible
Authentication	Login/register flows, validation	Ruoqi Hu
Vocabulary	Quiz logic, scoring accuracy	Weike Jin
Listening	Audio playback, question flow	Ruoqi Hu
Reading	Text display, question logic	Weike Jin

Module	Test Focus	Responsible
Writing	Input handling, save functionality	Ruoqi Hu
Speaking	Recording feature, playback	Weike Jin
Progress	Data accuracy, visualisation	Ruoqi Hu

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12. Deliverables

12.1 Sprint Deliverables

Sprint	PO	Deliverables
Sprint 1	Yanbin Xu	Auth system, Vocabulary module, Listening module
Sprint 2	Junfan Zhou	Reading module, Writing module
Sprint 3	Ruoqi Hu	Speaking/Presentation module, Progress dashboard
Sprint 4	Chenxi Huang	Integrated app, Documentation, Final presentation

12.2 Final Deliverables

Deliverable	Format	Owner	Due
Source Code	GitHub Repository	Yupei Yang	Week 6
Working Application	Deployed URL / APK	Yupei Yang	Week 6
Product Backlog	Trello/Document	Final PO (Chenxi)	Week 6
Technical Documentation	Markdown	Yupei Yang	Week 6
User Guide	PDF	Chenxi Huang	Week 6
Sprint Reports (x4)	PDF	Each Sprint's SM	Weekly
Final Presentation	PowerPoint	Final PO (Chenxi)	Week 6